

MALBORK, Poland

WMO: 121540

Lat: 54.027N

Lon: 19.134E

Elev: 5

StdP: 101.26

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.6	-13.0	-19.2	0.7	-16.2	-15.9	0.9	-11.9	13.3	6.0	11.5	3.9	1.4	70	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	29.3	20.0	27.4	19.1	25.8	18.4	21.5	26.9	20.4	25.2	19.4	23.9	3.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	14.5	24.0	18.8	13.6	22.7	17.7	12.7	21.4	62.5	27.0	58.6	25.2	55.2	23.9	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.7	7.7	-20.0	33.2	5.6	2.7	-24.0	35.1	-27.2	36.7	-30.4	38.2	-34.4	40.2		
(5)			-20.0	23.4	5.7	1.4	-24.1	24.4	-27.4	25.2	-30.6	25.9	-34.7	26.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-2.2	-0.9	2.8	7.9	13.1	16.5	19.0	18.5	14.4	9.1	4.6	1.1		
	DBStd	8.29	5.89	5.29	4.56	3.73	3.84	3.24	2.87	2.70	2.88	3.52	3.24	4.72		
	HDD10.0	1502	378	304	224	83	12	0	0	0	2	61	162	276		
	HDD18.3	3623	636	537	481	313	170	73	27	31	121	288	411	534		
	CDD10.0	1034	0	0	2	20	108	196	278	263	136	32	1	0		
	CDD18.3	112	0	0	0	0	8	18	46	36	4	0	0	0		
	CDH23.3	1034	0	0	0	7	84	196	388	314	44	0	0	0		
	CDH26.7	253	0	0	0	1	12	47	104	81	9	0	0	0		
(14) Wind	WSAvg	3.6	4.1	3.8	3.8	3.5	3.1	3.1	3.1	3.2	3.2	3.6	3.9	4.3		
(15) Precipitation	PrecAvg	638	37	33	34	34	58	64	80	76	67	63	50	42		
	PrecMax	796	77	60	66	80	91	122	158	149	175	129	105	75		
	PrecMin	524	11	16	10	2	22	8	15	19	20	12	14	17		
	PrecStd	70	17	10	15	20	21	32	34	34	38	34	22	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.9	10.0	16.9	23.9	28.2	31.1	32.1	31.9	27.8	20.1	12.6	10.9		
		MCWB	7.8	7.7	10.6	14.0	18.2	21.9	21.0	20.8	19.2	15.1	10.5	9.3		
	2%	DB	7.8	8.5	13.5	20.3	25.9	27.7	29.3	28.8	24.2	17.7	11.3	9.1		
		MCWB	6.2	6.4	8.3	12.8	18.3	19.5	20.2	19.8	16.9	13.9	9.7	7.7		
	5%	DB	6.3	6.8	11.2	17.5	23.2	25.4	27.2	26.7	22.0	16.0	10.1	7.9		
		MCWB	4.9	4.8	7.5	10.7	16.3	17.7	19.4	18.6	15.7	13.2	8.8	6.6		
	10%	DB	4.8	5.2	9.2	15.1	20.8	23.4	25.3	24.7	20.1	14.2	9.1	6.8		
		MCWB	3.8	3.7	6.3	9.6	14.9	16.7	18.8	17.8	14.9	11.9	8.0	5.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	8.3	11.5	16.0	19.9	23.1	23.0	22.4	19.6	16.1	11.2	9.6		
		MCDB	9.6	9.5	15.8	22.9	25.5	29.8	28.4	28.9	25.4	18.9	12.0	10.7		
	2%	WB	6.5	6.4	9.4	13.2	18.3	20.7	21.7	20.8	18.0	14.5	10.1	8.0		
		MCDB	7.6	8.2	12.0	18.6	24.5	26.0	26.7	26.0	22.7	16.9	11.1	9.0		
	5%	WB	5.0	5.0	7.9	11.5	17.0	19.0	20.7	19.7	16.6	13.3	9.0	6.6		
		MCDB	6.1	6.4	10.5	16.5	22.4	23.6	25.3	24.2	20.0	15.5	9.8	7.6		
	10%	WB	3.7	3.7	6.4	10.2	15.5	17.5	19.6	18.8	15.5	12.3	8.1	5.5		
		MCDB	4.8	5.1	9.0	14.1	19.6	21.5	23.9	23.0	19.1	14.1	9.0	6.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.8	5.7	8.0	10.9	10.9	10.8	10.0	10.4	9.2	7.1	4.5	4.2		
		MCDBR	4.4	6.6	11.5	15.6	14.1	14.4	13.5	13.8	12.9	9.6	5.6	5.0		
	5% WB	MCWBR	3.9	5.1	7.3	8.8	7.2	7.2	6.0	6.1	6.6	6.2	4.7	4.4		
		MCWBR	4.3	5.9	10.1	13.6	13.0	13.1	11.6	11.7	11.1	8.6	5.3	4.8		
(40) Clear-Sky Solar Irradiance	taub		0.307	0.338	0.369	0.416	0.406	0.402	0.436	0.424	0.381	0.362	0.346	0.307		
		taud	2.243	2.255	2.256	2.179	2.266	2.306	2.234	2.271	2.370	2.410	2.318	2.270		
	Ebn at Noon		634	740	800	807	841	847	807	793	784	712	584	556		
		Edh at Noon	61	85	105	129	125	122	129	117	94	73	59	50		
(44) All-Sky Solar Radiation	RadAvg		0.53	1.22	2.45	4.17	5.35	5.58	5.05	4.37	3.12	1.63	0.62	0.38		
	RadStd		0.08	0.18	0.31	0.52	0.53	0.49	0.58	0.42	0.36	0.25	0.08	0.05		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	-140	+70	N/A

Nomenclature: See separate page